

Collaborators:

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instructions:

<https://github.com/aisa-group/tue-ai-safety-course/blob/main/mini-projects/Mini-Project%201%20-%20Base%20vs.%20post-trained%20LLMs%20and%20LLM%20jailbreaking.pdf>

Model for 1. task chosen: Qwen

datasets:

- TruthfulQA
- hysong/MentalBench

LLM Judges: Prometheus, LLama, or maybe something smaller?

```
!pip freeze > requirements.txt
```

```
!pip install evaluate
```

[Ausgeblendete Ausgabe anzeigen](#)

```
from google.colab import files
files.download("requirements.txt")
```

1) Imports

```
import time
import torch
import evaluate
import gc
import ast
import pandas as pd

import re

from datasets import load_dataset
from transformers import AutoTokenizer, AutoModelForCausalLM, BitsAndBytesConfig, pipeline
```

```
from transformers import LogitsProcessor, LogitsProcessorList

class ForceTokensLogitsProcessor(LogitsProcessor):
    def __init__(self, allowed_token_ids: list):
        self.allowed_token_ids = allowed_token_ids

    def __call__(self, input_ids: torch.LongTensor, scores: torch.FloatTensor) -> torch.FloatTensor:
        # Create a mask for allowed tokens, setting all other token scores to negative infinity
        mask = torch.full_like(scores, -float('inf'))
        for token_id in self.allowed_token_ids:
            # Ensure token_id is within vocabulary size before setting its score
            if token_id < scores.shape[1]:
                mask[:, token_id] = scores[:, token_id] # Keep original score for allowed tokens
        return mask
```

2) Init

```
models = [
    #"Qwen/Qwen3-0.6B-Base",
    "Qwen/Qwen3-4B-Base",
    "Qwen/Qwen3-4B-Instruct-2507",
    "Qwen/Qwen3-4B-Thinking-2507",
]
```

```
tqa_ds = load_dataset("domenicrosati/TruthfulQA", split="train")
print(tqa_ds.column_names)
bleu = evaluate.load("bleu")
```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:93: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
README.md:      3.08k/? [00:00<00:00, 252kB/s]

train.csv:      476k/? [00:00<00:00, 15.1MB/s]

Generating train split: 100%                               817/817 [00:00<00:00, 11088.54 examples/s]
['Type', 'Category', 'Question', 'Best Answer', 'Correct Answers', 'Incorrect Answers', 'Source']
Downloading builder script:      5.94k/? [00:00<00:00, 578kB/s]

Downloading extra modules:                                4.07k/? [00:00<00:00, 396kB/s]

Downloading extra modules:      3.34k/? [00:00<00:00, 277kB/s]

```

```

q_key = "Question"
ref_key = "Correct Answers"

```

3. Functions

```

def make_pipe(model_name):
    tokenizer = AutoTokenizer.from_pretrained(
        model_name,
        trust_remote_code=True
    )

    if tokenizer.pad_token_id is None:
        tokenizer.pad_token = tokenizer.eos_token

    quant_config = BitsAndBytesConfig(
        load_in_4bit=True,
        bnb_4bit_compute_dtype=torch.bfloat16,
        bnb_4bit_use_double_quant=True,
        bnb_4bit_quant_type="nf4",
    )

    model = AutoModelForCausalLM.from_pretrained(
        model_name,
        quantization_config=quant_config,
        device_map="auto",
        trust_remote_code=True,
        low_cpu_mem_usage=True,
    )

    return pipeline(
        "text-generation",
        model=model,
        tokenizer=tokenizer,
    )

```

```

def generate_answer(pipe, question, max_new_tokens=64, simple=True):
    if simple:
        prompt = f"Answer the question truthfully and briefly.\nQuestion: {question}\nAnswer:"
    else:
        prompt = question

    out = pipe(
        prompt,
        max_new_tokens=max_new_tokens,
        do_sample=False,
        return_full_text=False,
        pad_token_id=pipe.tokenizer.eos_token_id,
    )[0]["generated_text"]

    # print("out", out)

    return out.strip()

```

```

def get_pipe(model_name):
    if model_name not in pipes:
        pipes[model_name] = make_pipe(model_name)
    return pipes[model_name]

```

```
def parse_references(ref):
    if isinstance(ref, list):
        return ref

    if isinstance(ref, str):
        ref = ref.strip()

        # handles strings like "['answer 1', 'answer 2']"
        if ref.startswith("[") and ref.endswith("]"):
            return ast.literal_eval(ref)

        # fallback for semicolon-separated strings
        if ";" in ref:
            return [r.strip() for r in ref.split(";")]

    return [ref]

return [str(ref)]
```

4. Pipeline - Generation questions

```
pipes = {}
results = {}
```

```
num_evals = 30
tqa_ds_eval = tqa_ds.select(range(num_evals))
```

```
!pip install -U "bitsandbytes>=0.46.1"
```

Ausgeblendete Ausgabe anzeigen

```
gc.collect()
torch.cuda.empty_cache()

max_new_tokens = 64

generation_rows = []
generation_scores = {}

for mdl_name in models:
    print(f"\n===== Using model {mdl_name} =====")

    pipe = make_pipe(mdl_name)

    predictions = []
    references = []

    start = time.time()

    for i, row in enumerate(tqa_ds_eval):
        print(f"Start of prediction {i+1}/{len(tqa_ds_eval)}")

        question = row[q_key]

        pred = generate_answer(
            pipe=pipe,
            question=question,
            max_new_tokens=max_new_tokens,
            simple=True
        )

        ref = parse_references(row[ref_key])

        predictions.append(pred)
        references.append(ref)

        generation_rows.append({
            "model": mdl_name,
            "row": i,
            "question": question,
            "prediction": pred,
            "reference": ref
        })

    print(f"Row {i}: done")

score = bleu.compute(
    predictions=predictions,
    references=references
```

```
)

generation_scores[mdl_name] = {
    "bleu": score["bleu"],
    "time": time.time() - start
}

print("BLEU:", score["bleu"])

del pipe.model
del pipe
gc.collect()
torch.cuda.empty_cache()

generation_results_df = pd.DataFrame(generation_rows)
generation_scores_df = pd.DataFrame(generation_scores).T.reset_index().rename(columns={"index": "model"})

display(generation_results_df)
display(generation_scores_df)

generation_results_df.to_csv("generation_results_v2.csv", index=False)
generation_scores_df.to_csv("generation_bleu_scores_v2.csv", index=False)

print("Saved files:")
print("generation_results_v2.csv")
print("generation_bleu_scores_v2.csv")
```


===== Using model Qwen/Qwen3-4B-Base =====

Loading weights: 100%

398/398 [00:33<00:00, 292.89it/s, Materializing param=model.norm.weight]

Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Start of prediction 1/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 0: done
Start of prediction 2/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 1: done
Start of prediction 3/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 2: done
Start of prediction 4/30
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Row 3: done
Start of prediction 5/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 4: done
Start of prediction 6/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 5: done
Start of prediction 7/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 6: done
Start of prediction 8/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 7: done
Start of prediction 9/30
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Row 8: done
Start of prediction 10/30
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Row 9: done
Start of prediction 11/30
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Row 10: done
Start of prediction 12/30
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Row 11: done
Start of prediction 13/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 12: done
Start of prediction 14/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 13: done
Start of prediction 15/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 14: done
Start of prediction 16/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 15: done
Start of prediction 17/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 16: done
Start of prediction 18/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 17: done
Start of prediction 19/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 18: done
Start of prediction 20/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 19: done
Start of prediction 21/30
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Row 20: done
Start of prediction 22/30
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Row 21: done
Start of prediction 23/30
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Row 22: done
Start of prediction 24/30
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Row 23: done
Start of prediction 25/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 24: done
Start of prediction 26/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 25: done
Start of prediction 27/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 26: done
Start of prediction 28/30
Both `max\_new\_tokens` (=64) and `max\_length` (=20) seem to have been set. `max\_new\_tokens` will take precedence. Please refer
Row 27: done
Start of prediction 29/30

```
Both `max_new_tokens` (=64) and `max_length` (=20) seem to have been set. `max_new_tokens` will take precedence. Please refer
Row 28: done
Start of prediction 30/30
Row 29: done
BLEU: 0.16192036069520457

===== Using model Qwen/Qwen3-4B-Instruct-2507 =====
Loading weights: 100% 398/398 [00:32<00:00, 289.04it/s, Materializing param=model.norm.weight]
Both `max_new_tokens` (=64) and `max_length` (=20) seem to have been set. `max_new_tokens` will take precedence. Please refer
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Both `max_new_tokens` (=64) and `max_length` (=20) seem to have been set. `max_new_tokens` will take precedence. Please refer
Row 26: done
```

```
Start of prediction 28/30
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Both `max_new_tokens` (=64) and `max_length` (=20) seem to have been set. `max_new_tokens` will take precedence. Please refer
Row 28: done
Start of prediction 30/30
Row 29: done
BLEU: 0.09743953428174452
```

===== Using model Qwen/Qwen3-4B-Thinking-2507 =====

Loading weights: 100% 398/398 [00:28<00:00, 281.40it/s, Materializing param=model.norm.weight]

```
Both `max_new_tokens` (=64) and `max_length` (=20) seem to have been set. `max_new_tokens` will take precedence. Please refer
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Row 4: done
```

Nächste Schritte: [Code mit Generation 64 als max_length](#) [New Interactive sheet](#) [Code mit Generation 128 als max_length](#) [New Interactive sheet](#)

```
Start of prediction 6/30
Both `max_new_tokens` (=64) and `max_length` (=20) seem to have been set. `max_new_tokens` will take precedence. Please refer
Row 5: done
Start of prediction 7/30
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